GEOMETRIC UNIBRANCHEDNESS DESCENDS ALONG FLAT MORPHISMS AT A POINT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one Morphisms of Spaces lemma and one Algebra lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Flatness at a point and \'etale local rings. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-at-point-etale-local-rings. Supplies flatness of the map of \'etale local rings.
- Going down for flat ring maps. Stacks Project, Algebra, Lemma algebra-lemma-flat-going-down. Supplies the transfer of the unique minimal prime.

Source reference. Original reference 04NG: sources/stacks-chow/tags.txt, source lines 6079–6079. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00HS: sources/stacks-chow/tags.txt, source lines 654–654. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $x \in |X|$. If f is flat at x and X is geometrically unibranch at x, then Y is geometrically unibranch at $f(x)$.*

Proof. Consider the map of étale local rings $\mathcal{O}_{Y,f(\overline{x})} \to \mathcal{O}_{X,\overline{x}}$. By Morphisms of Spaces, Lemma 04NG this is flat. Hence if $\mathcal{O}_{X,\overline{x}}$ has a unique minimal prime, so does $\mathcal{O}_{Y,f(\overline{x})}$ (by going down, see Algebra, Lemma 00HS). □

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